Midterm Sample: Solutions

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

1. (5 pts)

   (a) (1 pt) *Is f a valid function?* **Yes**: a function may map two values of x to the same value of y; what it cannot do is map one x to two values of y.

   (b) (1 pt) $A \cup B =$ Set of all integers.

   (c) (1 pt) $A \cup B = A \qquad A \cap B = B$

   (d) (1 pt) No: since the function is not one-to-one (it is not monotonic), the inverse does not exist.

   (e) (1 pt) $(0, 2)$

2. (5 pts) $A = I - X(X^T X)^{-1} X^T$

   (a) (3 pts) Say the dimension of X is $m \times n$. Then the dimension of $X^T_{n\times m} X_{m\times n}$ is $n \times n$, so the dimension of $(X^T X)^{-1}$ is also $n \times n$. This implies that the dimension of $X_{m\times n}(X^T X)^{-1}_{n\times n} X^T_{n\times m}$ is $m \times m$. Hence A is square (as is $X^T X$), but X need not be square.

   (b) (2 pts)

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\begin{aligned}
AA &= (I - X(X^T X)^{-1} X^T)(I - X(X^T X)^{-1} X^T) \\
&= I - X(X^T X)^{-1} X^T - X(X^T X)^{-1} X^T + X \underbrace{(X^T X)^{-1} X^T X}_{I} (X^T X)^{-1} X^T \\
&= I - X(X^T X)^{-1} X^T = A
\end{aligned}
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3. (10 pts)

   (a) (1 pt) $A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad v = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ 12 \\ 24 \end{bmatrix}$

(b) (3 pts) We first need to calculate all the cofactors of A.

$$|C_{11}| = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0 \quad |C_{12}| = -1\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1 \quad |C_{13}| = \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1$$

$$|C_{21}| = -1\begin{vmatrix} 0 & -2 \\ 1 & 1 \end{vmatrix} = -2 \quad |C_{22}| = \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} = 3 \quad |C_{23}| = -1\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = -1$$

$$|C_{31}| = \begin{vmatrix} 0 & -2 \\ 1 & 1 \end{vmatrix} = 2 \quad |C_{32}| = -1\begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix} = -1 \quad |C_{33}| = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$adjA = \begin{bmatrix} 0 & -2 & 2 \\ 1 & 3 & -1 \\ -1 & -1 & 1 \end{bmatrix}$$

(c) (2 pts)

$$|A| = a_{11}\,|C_{11}| + a_{12}\,|C_{12}| + a_{13}\,|C_{13}| = 1 \times 0 + 0 \times 1 + (-2) \times (-1) = 2$$

A is nonsingular as $|A| \neq 0$.

(d) (1 pt) Premultiplying by A^{-1}: $A^{-1}Av = A^{-1}b$. Since $A^{-1}A = I$, we have $v^* = A^{-1}b$.

(e) (3 pts) Since $A^{-1} = \frac{1}{|A|}adjA$,

$$v^* = \frac{1}{2}\begin{bmatrix} 0 & -2 & 2 \\ 1 & 3 & -1 \\ -1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 12 \\ 24 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -24 + 48 \\ 2 + 36 - 24 \\ -2 - 12 + 24 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \\ 5 \end{bmatrix}$$

Checking: $12 - 2(5) = 2$, $7 + 5 = 12$, $12 + 7 + 5 = 24$. So agriculture produces 12, manufacturing 7, and services 5 (billions of dollars).